

GROUPS, BOUNDARY SPANNING, AND THE TEMPORAL IMAGINATION

Allen C. Bluedorn and Rhetta L. Standifer

ABSTRACT

The temporal imagination is the understanding of the intersection of one entity's timescape with the larger timescapes of which that entity is a part. We examine in detail what the temporal imagination is, complemented with a discussion of the related timescape idea, and why the temporal imagination is necessary to function in any timescape. We also discuss group attributes that will likely affect the development of the temporal imagination and its use and how its use in group boundary spanning efforts affect both the groups and the larger organization.

INTRODUCTION

The temporal imagination is the ability to understand the intersection of one entity's timescape with the larger timescapes of which that entity is a part. The entity can be an individual, a group, or an entire organization, but this chapter focuses on boundary spanning groups and their parent organizations. Several attributes of boundary spanning groups that are likely to influence the quality of the group's temporal imagination are then presented and analyzed (group diversity, shared mental models within the group, longevity, and autonomy).

Two generations ago, C. Wright Mills' wrote *The Sociological Imagination* (1959), which ever since has provided an enduring description of the enterprise

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of social science as defined by the concept in the book's title. According to Mills, "The sociological imagination enables us to grasp history and biography and the relations between the two in society" (p. 6). Further, the sociological imagination allows people "to understand what is happening in themselves as minute points of the intersection of biography and history in society" (p. 7). Thus it would be fair to say that the sociological imagination is the ability to understand the intersection of history and biography, and in that understanding to comprehend the effect each has on the other.

It is impossible, however, to understand that intersection completely without an understanding of the temporal dimensions of both biography and history, such an understanding being the proper focus of the *temporal imagination*. The term "temporal imagination" has been used sparingly in a variety of disciplines, including literature (Barth, 1991; Weitzel, 1999), sociology (Nock, 1998), and political science (Chowers, 1998). To the best of our knowledge, it has not been used previously in the organization sciences.

What is the temporal imagination? Why is it necessary to function in any timescape? How does its use in group boundary spanning efforts affect both the groups and the larger organization? These will be the fundamental questions addressed and developed in this chapter. We will begin by considering the concept of the temporal imagination itself, complemented with a discussion of the related timescape idea, followed by a discussion of group attributes that will likely affect the development of the temporal imagination and its use.

THE TEMPORAL IMAGINATION

Cervantes died on April 23, 1616. Shakespeare also died on April 23, 1616, which leads without much thought to the conclusion that both bards died on the same day – an erroneous conclusion. For the date of Cervantes' death is the date according to the Gregorian calendar, which came into use in Spain in 1582 (Richards, 1998, p. 248), whereas the date of Shakespeare's passing is the date according to the Julian calendar (details about Cervantes and Shakespeare from Whitrow, 1988, p. 119), which was still in use in England at the time and would continue to be used there until 1752 (Richards, 1998, p. 248). Thus Shakespeare actually lived ten days longer than Cervantes because the date of Shakespeare's death on the Gregorian calendar would be May 3, 1616 (Whitrow, 1988, p. 119).

The Gregorian calendar was instituted by Pope Gregory XIII in 1582 because the Julian calendar was falling out of step with the seasons, leading Easter to head into parts of the year the church felt were undesirable (Bluedorn, 2002, pp. 160–161). However, Gregory was a pope, so Anglican England did not adopt the

reformed calendar until 1752, thereby creating the false impression that Cervantes and Shakespeare died on the same day.

In this example, it would be fair to say that anyone who saw the two dates and concluded that these two giants of Western literature had perished on the same day was using a limited temporal imagination. To reach the proper conclusion about their two deaths, at a minimum one would need to know the following: (1) the calendar used throughout much of the world in the 21st century for secular matters was instituted in the late sixteenth century (1582); but (2) its adoption was a slow process, especially outside Catholic countries, and that process of adoption would continue well into the twentieth century. In particular, one would need to know that England did not adopt the Gregorian calendar until 1752, well after Shakespeare's death. Without this knowledge, it would be natural to assume that the dates given for both men's deaths were dates on the same calendar, hence that both deaths had occurred on the same day. For time and things temporal tend to be cultural matters, and they often reside in the deepest parts of culture, at the level Schein labeled basic underlying assumptions (1992, pp. 16–22). As such, it is easy to take physical manifestations of culture, like the calendar, for granted and to assume that they have always been in use in their current forms.

The calendar – be it Gregorian, Julian, Hebrew, Chinese, or Mayan – serves as an example of one component of a *timescape*. Although others had used the term earlier (e.g. Benford, 1980/1992), Barbara Adam (1998) developed the concept of timescape, writing:

Where other scapes such as landscapes, cityscapes and seascapes mark the spatial features of past and present activities and interactions of organisms and matter, timescapes emphasise their rhythmicities, their timings and tempos, their changes and contingencies. A timescape perspective stresses the temporal features of living. Through timescapes, contextual temporal practices become tangible. Timescapes are thus the embodiment of practiced approaches to time (p. 11).

The calendar and calendar system that a society uses are one part of that society's "practiced approaches to time," hence a part of its timescape. When such practices are combined mindfully with a broader understanding of the timescape to consciously affect decisions and actions, it is appropriate to say that the temporal imagination is being employed. Thus we define the temporal imagination – a derivative of the sociological imagination (understanding the intersection of history and biography) – as *the ability to understand the intersection of one's own temporal behavior (i.e. timescape) with that of the larger timescape*.

Temporal imaginations promote the functioning and survival of individuals and groups. At times, such an imagination, if used properly, can provide a competitive advantage leading to particularly high performance. Given such potential impacts, developing a strong temporal imagination seems imperative. Developing a

powerful temporal imagination begins by learning what characteristics and dimensions to look for in other timescapes – as well as in one’s own. Thus, a necessary condition for developing an insightful temporal imagination is the ability to discern a timescape’s temporal dimensions. Toward the end of promoting such discernment, we identify six universal dimensions of timespaces in the following discussion.

EXPLORING THE TIMESCAPE

We believe that every timescape will have its unique attributes, if for no other reason than that every socio-cultural phenomenon will have a unique history, hence its development will have produced idiosyncratic forms and characteristics. Nevertheless, there are fundamental temporal phenomena that will occur in any timescape. The ability to identify these phenomena and understand their roles in the specified timescape are requisite abilities for a well-functioning temporal imagination.

Adam suggests several fundamental temporal phenomena in her general description of timescapes, emphasizing “their rhythmicities, their timings and tempos, their changes and contingencies” (1998, p. 11). Rhythmicities and their timings and tempos are important temporal phenomena that combine to form portions of a timescape, as is indicated in Adam’s terms “timings” and “contingencies.” [Bluedorn \(2002\)](#) identified some of the same fundamental temporal phenomena that Adam suggested, also emphasizing rhythms and the relationships among rhythms.

Both researchers also identified how rapidly things are done as an important temporal characteristic, albeit Adam uses the term “tempo”; Bluedorn, “speed.” Bluedorn identified polychronicity, temporal depth, temporal focus, and punctuality as fundamental temporal phenomena as well.

Polychronicity is the extent to which people: (1) prefer to be engaged in two or more tasks or events simultaneously and are actually so engaged (the preference strongly implying the behavior and vice versa); and (2) believe their preference is the best way to do things ([Bluedorn, 2002](#), p. 51). Anthropologist Edward Hall developed the concept of polychronicity and was the first to use it to explain human behavior (see [Bluedorn, 1998](#); [Hall, 1959/1981, 1983](#)). Hall used the concept to describe and explain both individual and cultural differences, and [Bluedorn \(2002, p. 48\)](#) described it as “a fundamental strategy for engaging life.” Polychronicity is an essential element in any well-developed temporal imagination because it describes a fundamental temporal characteristic of individuals and their timescapes as well as of a collectivity such as an organization and its timescape.¹ Thus those seeking to develop an insightful temporal imagination should certainly examine the

timescape to note its level of polychronicity – just as people developing a temporal imagination should note the timescape’s characteristic temporal depth and focus.

Social scientists have identified “time orientation” (e.g. Kluckhohn & Strodtbeck, 1961, pp. 13–15) as a fundamental way to describe how individuals and cultures differ in their beliefs and emphases on time, the term tending to refer to the relative emphasis placed on the past, present, and future (see Bluedorn, Kaufman & Lane, 1992, p. 26). More recently Bluedorn (2000, 2002) has pointed out that this single concept often masks two independent constructs: one having to do with which general temporal region (past, present, or future) a person or culture tends to emphasize; the other having to do with how far into the past and the future people and cultures tend to look when they are thinking about things. The former he termed temporal focus; the latter, temporal depth. Therefore, we suggest that these are important temporal aspects of entities and the timescapes in which they reside. They are best considered as two distinct temporal phenomena rather than one.

Finally, following the work of Levine (e.g. Levine, 1997; Levine, West & Reis, 1980; Levine & Wolff, 1985), Bluedorn (2002) indicated that punctuality – how important it is to be on time and what is defined as being on time – is a fundamental temporal phenomenon, and as such, like the other temporal phenomena inventoried so far, it can properly describe both entities and their timescapes.

To summarize, we have identified six fundamental temporal characteristics that would seem to be properties of all timescapes: rhythmicities/rhythms, tempo/speed, polychronicity, temporal depth, temporal focus, and punctuality. Given that they are reasonably well established in the time literature as such, they are likely to be found in all timescapes, albeit other temporal characteristics are likely to be found in timescapes too. Further, the temporal imagination can now be recast as *understanding the intersection of two or more timescapes* because the entity, including individuals, will have regular practiced approaches to time, thus allowing us to speak of an individual, group, or organization’s *internal* timescape as well as of the timescapes of the larger systems of which the entities are a part. Knowing, however, which fundamental temporal properties a timescape possesses is not enough to fruitfully employ the temporal imagination; other knowledge and behaviors are necessary too.

DEVELOPING THE TEMPORAL IMAGINATION IN GROUPS

To develop and use the temporal imagination effectively, one must identify and develop knowledge about one’s internal timescape as well as the larger timescapes

of which one is a part. For groups and organizations to do so, knowledge is required about the potential advantages of groups and of how groups and other collectivities interact with their environments.

Groups Advantages

Given finite time and other limited resources, the individuals who compose a group should be able to see more aspects of a timescape's reality than any solitary individual, *ceteris paribus*. Further, such an advantage will accrue only when the members of the group attempt to process their knowledge to build a more accurate and larger overall picture of that timescape reality, rather than when the group's members argue that their observations are the only correct descriptions of the timescape. The group will need to see its goal as producing a larger, more accurate and complete description of the timescape, but a description that is always subject to revision and additional development. Thus, the argument should never be about issues such as: Is the elephant more like a wall, a spear, a snake, a tree, a fan, or a rope? – the debate carried on by the six blind men in [John Saxe's \(1892\)](#) famous poem. Rather, the discussion of how to interpret the observations should focus on what information each observation provides about the reality in question. To get the group to operate in this fashion and with this orientation is where knowledge of organizational behavior and group dynamics can truly inform this task, as will be demonstrated shortly. But before exploring further the inner workings of groups that might enhance their ability to use temporal imaginations, the relationship between groups and other collectivities with their environments needs to be addressed, especially one strategic aspect of that relationship.

Actors are Parts of Their Own Timescapes

[Gareth Morgan \(1997, p. 258\)](#) has noted, “Many organizations encounter great problems in dealing with the wider world because they do not recognize how they are part of their environment.” Implicit in our analysis is the premise that the larger timescapes that a group or organization participates in are crucial aspects of their environments. A critical – albeit subtle – point for any temporal imagination to recognize is that the actor itself is a part of the larger timescape in which it exists, which means that to understand the intersection of the actor with its timescape requires much more comprehension than simply noting the fundamental temporal characteristics of both the actor and the timescape and how they are similar and different.

As basic as these points are to an adequate temporal imagination, they form but the foundation upon which truly imaginative temporal insights are based. Once these basics are understood, the next step is to proceed to a deeper understanding of the intersection of actor (e.g. individual, group, or organization) and timescape. Namely, how do specific actions that actors take affect/change the timescape, and in turn, how do those subsequent timescape changes affect the actor? [Morgan \(1997, p. 260\)](#) provides examples of how organizations, sometimes acting as an entire industry, make decisions and take actions with only the short-term future in mind. Considering only short-term consequences, this implies, however, that had longer-term futures and other possible impacts on both the environment and their consequences for the organization been considered, other decisions might have resulted.

If we consider an example focusing on the temporal aspects of both the organization and its timescape, we may see the type of thinking the temporal imagination might employ to identify such effects and consequences. The example is the development of the emphasis on speed that is at the heart of the organizational strategy known as time-based competition ([Blackburn, 1991](#); [Stalk & Hout, 1990](#)). Let us consider: (1) this strategy from the perspective of a single focal organization; (2) the impact this organization might have on its timescape if it adopts this strategy; and (3) the likely subsequent impacts these timescape changes might have on the organization itself.

Assume that the organization's strategic management team is considering the option of time-based competition, that is, of altering its way of operating so that it can develop, produce, and bring to market new products faster than its competitors (see [Stalk & Hout, 1990, p. 29](#)). Making these changes will create costs for the organization as well as potential benefits, so the strategists would need to consider the ratio of the two, as well as the feasibility of making the changes and the probability that the strategy will work (i.e. will produce the minimal acceptable benefits and do so with an acceptable cost-benefit ratio). Such traditional thinking is likely to focus on the results for the organization, and to focus on them in terms of profitability.

Let us now add thinking directed by the temporal imagination to this decision and see some of the issues it might highlight. First, what are some of the organizational changes regarding temporal matters that a shift to a time-based strategy might require or produce? Time-based strategies require, virtually by definition, an emphasis on speed, on doing things faster. Thus one of the immediate challenges for the organization will be a change in the temporal portion of its culture, of its internal timescape, to value and practice doing things faster. This will be necessary because no strategy can be implemented optimally when it is incompatible with the organization's culture ([Bluedorn & Lundgren, 1993](#)). Clearly, this culture-change

effort will be easier the closer the organization's timescape already is to values and "practiced approaches" emphasizing the importance of doing things fast. This is an easy observation, one that does move beyond just basic descriptions of timescape features – the internal timescape in this case – but one that does not overly stretch the developing temporal imagination.

More challenging are the next two questions: (1) What effect will the shift to this strategy have on the larger timescape in which the organization functions? and (2) What effect might those changes to the larger timescape have on the organization? First, let us assume that the focal organization is an industry leader, or that if it was not before the change, that it became one after the shift to the time-based strategy. (We will beg the question of causality here and simply assume an empirical association with organizational success.) Institutional forces such as mimetic imitation (DiMaggio & Powell, 1983) will likely lead other organizations operating in the same timescape to adopt versions of time-based strategies themselves. Assuming they do, this will directly change at least one fundamental dimension of the timescape: the extent to which doing things fast is valued. Compared to the moments before the focal organization adopted the time-based strategy, the entire timescape will now place a much greater emphasis on speed, on doing things fast. Other temporal dimensions of the timescape are likely to be affected too, but those effects are more difficult to anticipate.

Second, the focal organization's original time-based strategy will no longer confer the same degree of competitive advantage it once did, if at this point it will confer any competitive advantages vis-à-vis the competition at all. This suggests the temporal imagination would reveal that the competitive advantage conferred by the initial version of the time-based strategy will be finite rather than indefinite, and that subsequent competitive advantages will have to come by either enhancing the time-based strategy, that is, by doing things even faster, or by seeking competitive advantages in other areas.

To follow through with this example, let us assume the focal organization's strategists decide to enhance the time-based strategy in response to the timescape's changes, that they make further changes so that their organization values and practices doing things even faster. As would be expected, the other organizations in the timescape would respond as they did before, by enhancing their own time-based strategies, and the entire timescape comes to value and practice doing things even faster still. But as only the most creative temporal imagination might anticipate from this spiral of positive feedback loops, at some point further enhancements could well become counterproductive, that things would be done *too* fast. For example, at one point in the history of the Japanese computer industry, product changes and improvements were being developed and brought to market so quickly that customers were delaying purchases in anticipation of yet better

machines being available later. This led to a conscious effort to actually reduce the rate of product development in this industry (Ancona & Chong, 1996, p. 262).

This example demonstrates that organizations are not only part of their own timescapes, but also how the temporal imagination is important. We will now see in more detail how that imagination can be developed and enhanced in a particular type of group.

GROUPS AND BOUNDARY SPANNING

We stated earlier that given the finite time and resources usually available to organizations, groups are better suited to the development of a robust temporal imagination than individuals because of the aggregated vision that comes from multiple perspectives. In this regard, all groups (and their organizations) benefit from this collective perspective; but particularly useful are those groups that engage in boundary spanning activities on behalf of their organizations.

Boundary spanners and boundary spanning groups establish and maintain relationships between the organization and external social networks, providing a primary conduit of information about these networks that the organization uses to make decisions. Therefore, groups engaged in boundary spanning activities interpret the external environment to the organization, as well as represent the organization back to entities in the environment. (This description is reminiscent of Likert's description of individuals performing linking functions between groups, 1961, p. 166, albeit we focus more on groups as the linking agents than individuals.) In the past, boundary spanning roles included sales representatives, recruiters, and bargaining agents (Kahn, Wolfe, Quinn, Snoek & Rosenthal, 1964). However, technology and increased interorganizational collaboration have broadened the definition of what constitutes a "boundary spanner."

Just as past researchers (March & Simon, 1958; Pennings, 1998; Thompson, 1967) have emphasized the importance of organizational boundaries and the boundary spanner's role as a conduit of information from a traditional perspective, we believe it is equally important to the temporal viewpoint. By gathering and assessing information from outside the organization and by using that information to aid the organization in its decision making, the boundary spanning group develops an understanding of its own internal timescape, that of its organization, and of the relationships of both to the larger timescapes in the environment. It also addresses the issues of how to "link" itself and the organization to the environment so that the environment and organization are each represented to the other, thereby enriching the group's and the organization's temporal imagination. Therefore, we will restrict our discussion of groups to those whose primary purpose is to perform

boundary spanning activities and part of whose mission is the development and understanding of the timescape and hence the temporal imagination of the group and the organization.

As mentioned previously, timescapes are formed by practiced approaches to time. For groups functioning within an organizational framework, these practiced approaches can include multiple calendars (society's, the fiscal year, etc.), deadlines imposed within (or upon) the group, meetings, the methods of synchronizing member schedules, seasonal cycles, and the various idiosyncrasies of the industry in which the group functions. The ways in which boundary spanning groups address these temporal practices can change, depending upon the attributes or characteristics of the group. In other words, the characteristics of a group affect the way in which that group confronts or handles practiced approaches to time, creating the group's unique timescape. Likewise, attributes of boundary spanning groups also affect the group's temporal imagination and its ability to understand and effectively use that imagination to help it and its organization manage their relationships with the larger environment.

Because of this, we shall now discuss how various group attributes can contribute to the internal timescapes of groups focused upon boundary spanning activities and the resultant effect on such groups' temporal imaginations. Although the general research on group attributes is both broad and extensive, we will focus on the following specific attributes: informational diversity within the group, temporal diversity within the group, extent of shared mental models within the group, group longevity, and degree of group autonomy.

Prior to a discussion concerning group diversity, it should be noted that researchers study many sources of diversity in groups, including gender, race, tenure, and educational backgrounds. In fact, past researchers (Guzzo & Dickson, 1996; Jackson, May & Whitney, 1995; Magjuka & Baldwin, 1991) have linked group diversity to group effectiveness, creativity, and decision-making. Given the vast scope of research on this topic, however, and with an eye toward what is particularly relevant to groups engaged in boundary spanning activities, we restrict our discussion to two types of diversity, one of which is found in the extant literature and one of our own making: informational diversity and temporal diversity, respectively.

Informational Diversity

Informational diversity is defined as differences in knowledge and perspectives among group members due to differences in education, experience, and expertise (Jehn, Northcraft & Neale, 1999). The nature and purpose of boundary spanning

groups, which often entail cross-functional group membership and/or different organizational origins (i.e. a group created for the purposes of developing an ongoing, interorganizational relationship), increases the likelihood of such diversity within these groups. The empirical research that examines the effects of such diversity within groups has produced mixed results (Ancona & Caldwell, 1992; Jackson, 1992). With regard to boundary spanning groups, however, we feel greater levels of informational diversity within such groups will provide an advantage over groups with lower levels of such diversity, in terms of the group's timescape and temporal imagination.²

One reason is that informational diversity creates a broader, richer group timescape for boundary spanning groups. This occurs because the diverse group of boundary spanning members gathers information and material from a greater number of sources as individual members target the areas with which they are familiar. Likewise, such diversity in group members provides a richer foundation of experience from which to assess the information and material gathered over the course of boundary spanning activities.

Groups with diverse experience can also better determine how the results of their boundary spanning activities can best serve the temporal needs of their own group and the organizations they represent. It is interesting to note that we (and researchers such as Ofori-Dankwa & Julian, 2001) question whether it is sometimes better *not* to be perfectly synched rhythmically in all ways with external components of one's larger social system. Indeed, Ofori-Dankwa and Julian (2001) have suggested the idea of an "entrainment quotient," meaning the ease and speed with which individuals, groups, and organizations can adjust time orientations to better assess the need to synchronize with outside components. Groups whose members' diversity of experience allows the groups' timescapes to effectively "link" to the timescape of its environment, and whose temporal imagination is such that the group is able to detect and effectively manipulate this link, might have a higher "quotient" than a more homogeneous group.

Later in this chapter, we will examine several caveats that are necessary when discussing the effects of diversity in groups. However, we prefer to discuss these notes of caution after our examination of the second type of group diversity: temporal diversity, to which we now turn.

Temporal Diversity

Just as diversity of experience affects a boundary spanning group's timescape and temporal imagination, the extent to which the group members differ in key temporal dimensions also influences the group's ability to understand and develop

its own internal timescape and how the group fits within the timescape of its organization and the larger environment. We define temporal diversity as the extent to which the group exhibits variance along the key temporal dimensions discussed earlier in the chapter. These temporal dimensions would include speed and punctuality, rhythmic activity, polychronicity, and temporal focus and depth.

In the previous section, we stated that informational diversity leads to positive outcomes (e.g. a well-developed timescape and temporal imagination). However, as a study conducted by [Jehn et al. \(1999\)](#) illustrates, the results of diversity are not always positive. As we will see, diversity is not uniformly beneficial with regard to these temporal dimensions either.

For example, we feel that in terms of speed and punctuality, groups exhibiting higher levels of diversity will experience more problems than groups that are more homogeneous. Hence, if some members emphasize fast decision-making while other members choose to deliberate for longer periods of time, conflict (i.e. process, task, and relationship conflict) is more likely. As a meta-analysis of conflict's effect on team performance shows ([De Dreu & Weingart, 2002](#)), such conflict among group members diminishes the performance (i.e. the quality of decisions) of the group. Likewise, if punctuality is terribly important to certain members, while others are more cavalier about arriving on time, similar results should be expected.

We suggest that the group's emphasis on speed and punctuality should be dictated by the cultural rhythms and the dynamism of the environment with which the group is scanning and interacting, rather than by the individual preferences of group members. Such sensitivity to the external timescape exemplifies the importance of a well-developed temporal imagination in boundary spanning groups, as well as the importance of understanding how the timescape of the organization and the group impacts the larger environmental timescape, and vice versa. If a group has the ability to detect rhythmic patterns and practices within its external environment, and if the group is able to effectively manage its own temporal behavior to its advantage, both the group and the organization(s) it represents benefit. For instance, [Eisenhardt \(1989\)](#) showed that dynamic environments dictate the need for faster speed in decision making. Therefore, a boundary spanning group that understands the intersection of its timescape with those in its environment will respect the cultural norms of that environment and respond appropriately with faster speed in its temporal behavior. And to optimally benefit its parent organization, the group will not only modify its own temporal behaviors to deal with those in environmental timescapes, but it will work to produce broader temporal adjustments from other relevant parts of the organization toward a similar end.

Another temporal dimension – rhythms – presents more of a theoretical challenge. Nevertheless, we feel that the ability to detect rhythms in the environment is essential to boundary spanning groups and is likely the *sine qua non* of their temporal imaginations. Therefore, although every group member need not be a master at environmental rhythm detection, this ability is essential to the group's ability to understand timescapes and thus exercise the group's temporal imagination effectively. Organizations can, of course, help promote a group's ability to detect and interpret environmental rhythms by consciously assigning individuals who exhibit this ability to boundary spanning groups (see [Bluedorn, 1997](#), about how to identify such individuals and for an analysis of how they detect environmental rhythms).

While the general ability to detect environmental rhythms is crucial for boundary spanning groups, we believe that boundary spanning groups whose members can detect different *types* of rhythmic patterns in the environment will be better suited to their task than groups whose members can detect only one type of rhythm. Therefore, diversity in the ability to detect different patterns is preferable among the members of boundary spanning groups. To better understand why, consider the painting by Georges Seurat, *A Sunday Afternoon on the Island of La Grande Jatte*, created using the technique of Pointillism. The portrait is made up of nothing but small dots of pure color; it is up to the eye of the viewer to mix and interpret the dots as a unified image. To truly understand the significance of Seurat's work, the person gathering the information must see and appreciate both the micro patterns of the dots as well as the unified pattern created by them.

So it is with boundary spanning groups interpreting the external timescape and relating it back to their parent organizations. This is because most temporal phenomena are either rhythms or properties of rhythms (e.g. polychronicity describes people's characteristic rhythms for engaging events and activities). And since rhythms are often subtle and can possess wave lengths with orders-of-magnitude differences (cf. [Bluedorn, 2002](#), pp. 130, 138), individuals who vary in their characteristic temporal depths will aid the group's ability as a whole to detect and interpret a wider variety of rhythms (i.e. rhythms whose wave lengths vary widely). Thus using Seurat's painting as a metaphor for the environment, group members who stand close to the painting (i.e. who have short-term temporal depths) will be most likely to see the dots and details about them in the painting, whereas members who stand back several feet (i.e. who have long-term temporal depths) will be more likely to see the images formed by the dots. So if boundary spanning groups can access and interpret different rhythmic patterns present in the environment and their organization, doing so will promote a broader, richer temporal imagination and timescape for the group, and ultimately for the group's parent organization.

In terms of polychronicity, boundary spanning groups exhibiting relatively polychronic behaviors can be more effective in their boundary spanning alignment activity than teams exhibiting relatively monochronic behavior; therefore, polychronic behavior by the group is preferred because more group polychronic behavior is likely to generate more information and environmental scanning, and accomplish a larger variety of tasks. Polychronic groups are more likely to see more aspects of the Seurat painting.

Does this mean that there should be little diversity among group members in their own polychronic/monochronic preference? In other words, should relatively monochronic people be discouraged from boundary spanning group membership? Not necessarily, because it is the group as a whole that should exhibit polychronic behavior with regard to the goal of boundary spanning. Thus a group with five very monochronic members, each working on a different task for the group, is by definition much more polychronic *at the group level* than another group with five very polychronic members each of whom is working on the same group task. Whether or not polychronic individuals can more readily work together in a boundary spanning group, especially on the tasks of integrating and interpreting the disparate data about external timescapes, remains an open question. It would be premature to specify the desired characteristics of individual group members in terms of their positions on the polychronicity continuum.

The final temporal dimension under discussion here is temporal focus. (We previously addressed the need for diversity of temporal depths in our discussion of the Seurat painting). Just as with informational diversity and the ability to detect different types of rhythms, diversity with regard to temporal focus is preferable for groups engaged in boundary spanning activities. If a group is charged with gathering and interpreting vast amounts of information from a variety of environmental sources, it is helpful if some members focus on the past and others on the future, thereby painting a more detailed, expansive portrait of the environment and how it relates to the organization.

Cautions and an Example

Earlier, we remarked that several cautionary notes are necessary when discussing the effects of diversity in groups. Recall for a moment the discussion of Seurat's painting and the problem of observing and interpreting the reality of it. The existence (and awareness of the existence) of multiple perspectives (e.g. different perspectives due to experience, temporal focus, etc.) can aid in a clearer and fuller vision of the painting. However, for these multiple perspectives to coalesce into a complete and accurate view of the painting, the members of the group must

be able (and willing) to communicate, combine, and arrange the assortment of perspectives in a meaningful way. What follows are several thoughts regarding the successful integration/utilization of multiple perspectives due to informational or temporal diversity among group members. It should be noted, however, that the successful integration of multiple perspectives in groups constitutes a large, formidable topic in and of itself. Therefore, we limit our discussion to a few suggestions particularly relevant to boundary spanning groups.

First, a diversity of perspectives creates the need for more integrating mechanisms to achieve better coordination (Lawrence & Lorsch, 1967). This could potentially make boundary spanning activities more complex for the group; however, the benefits outweigh the potential problems in this case. Promoting and maintaining a group atmosphere in which multiple perspectives are welcome should at least help reduce problems for such boundary spanning groups and increase the likelihood that such mechanisms will be encouraged and used.

Second, boundary spanning groups whose membership consists of a high degree of diversity with regard to organizational origin (e.g. cross-functional groups created for interorganizational purposes) should expect a decrease (albeit not completely) of positive results because of a lack of familiarity among members and possibly a lack of trust. One way to counteract this problem would be to allow such interorganizational groups to remain intact for longer periods of time, thereby allowing members to become familiar and comfortable with one another. (See our discussion below about group longevity for further elaboration on this point.)

Finally, we must ask: How much diversity is too much diversity? The answer is, *it depends*. For instance, greater levels of informational diversity are desirable for the development of a boundary spanning group's temporal imagination. In fact, a certain amount of informational diversity is necessary in boundary spanning groups for a rich temporal imagination to develop. However, moderate levels of diversity can cause groups to splinter into subgroups along "faultlines," causing a detrimental effect to the group and its purpose (Lau & Murnighan, 1998). Higher-than-moderate levels of informational diversity might result in better development than in groups with moderate levels of such diversity, particularly with regard to the tasks often associated with boundary spanning groups. (The strength of such an effect might depend upon the salience of differences relative to the task at hand, and whether these differences were perceived and subgroups created early in the formation of the group, Lau & Murnighan, 1998.)

The optimal level of *temporal* diversity, however, requires further consideration. As we discussed, temporal diversity involves multiple key dimensions; these dimensions can (and we expect, will) differ in terms of preferable levels of diversity. In terms of speed and/or punctuality, we expect moderate levels of diversity to

cause results similar to those of informational diversity. For example, if the group is split evenly between members who are always on time and members who are always late (i.e. two groups), a faultline might emerge between the “on-time” and “late” groups, causing conflict. (Of course, if the need to be punctual is not crucial to the task, then its salience might be minimized.) If, however, there is no clear division, either because of common punctuality practices (i.e. one group) or because there is a high level of diversity (i.e. many groups) in terms of punctuality, the issue of being on time might not emerge as an issue within the group because subgroups will not appear.

With regards to other temporal dimensions, however, we suggest that moderate levels of diversity might not have such a detrimental effect. For instance, in terms of temporal focus or the boundary spanning group’s ability to detect different types of rhythms, higher levels of this type of temporal diversity might have a positive effect up to a certain point. When a variety of perceptions is valuable to the task at hand (e.g. creativity and decision-making), researchers have linked diversity to positive effects in groups (Guzzo & Dickson, 1996; Jackson, 1992). We suggest that such is the case in terms of these particular types of temporal diversity. At extreme levels of such temporal diversity, however, the negative aspects will most likely outweigh any positive benefits due to increased conflict and uncertainty, such as has been suggested by other researchers (Greening & Johnson, 1997; Hambrick, Seung Cho & Chen, 1996; Williams & O’Reilly, 1998). Of course, there is little or no empirical evidence with regard to temporal diversity or the development of a group’s temporal imagination at this time; as such, our discussion is preliminary to such research.

Let us now consider an example of how these forms of diversity in boundary spanning groups might influence a group’s understanding of the synergistic relationship of its parent organization’s internal timescape with the larger timescape of the external environment. We will return to the example of time-based competition. A boundary spanning group whose members have differing areas of temporal expertise will be more likely to develop a clearer, more accurate composite of the internal and external timescapes. This is so because, at the most fundamental level, such a group will be better suited to assess temporal differences between the organization and the environment. The fact that the group can access and interpret information from a variety of environmental sources also allows it to determine how organizational changes will alter the timescape of the environment and how, in turn, those external alterations will impact the organizational timescape. In time-based competition there is a point of diminishing returns because of the detrimental, cyclical effects as both the organization and the environment respond to the other with an ever-greater emphasis on speed. A boundary spanning group that can gather information from a wider variety of environmental sources might

detect that effect faster than a group that only accesses one area of environmental information.

Likewise, differences along temporal dimensions (i.e. overall temporal diversity) affect how time-based competition might have been approached in the earlier example. For instance, a group that emphasizes speed might be able to determine the extent to which the environment embraces (or rejects) the idea of speed and whether a competitive advantage based on speed can be sustained by the organization within that environment.

Shared Group Mental Models

Another potentially influential attribute of the temporal imagination is the existence of shared mental models among group members. A “mental model” is defined as an organized knowledge structure that allows for effective interaction with the environment (Mathieu, Heffner, Goodwin, Salas & Cannon-Bowers, 2000). The idea of “shared” models of understanding dates back to the 1930s, when Mead (1934/1974) described the necessity of such shared models for teams engaged in complex activities requiring a good deal of cooperation. When a mental model is “shared,” teams are better able to effectively coordinate activity and decision making (Stout, Cannon-Bowers & Salas, 1996). Shared mental models are helpful to groups engaged in complex, dynamic tasks (Mathieu et al., 2000) and have been positively linked to organizational team effectiveness (Klimoski & Mohammed, 1994). Specifically, shared mental models make it possible for groups to explain phenomena, draw inferences, make predictions of future events, and recognize relationships among environmental elements and other contextual components (Cannon-Bowers, Salas & Converse, 1993). Given this, we feel shared mental models are particularly helpful to groups engaged in boundary spanning activities.

According to shared mental model researchers (e.g. Cannon-Bowers et al., 1993; Klimoski & Mohammed, 1994), teams can use different types of mental models to achieve collaborative, cooperative goals. These models include a shared understanding of technology, mutual knowledge about procedures or environmental conditions, and collective knowledge of member roles, sources of information, and common patterns of interaction.

We would like to add another shared mental model to this list: a shared mental model that denotes a common (shared) timescape and temporal imagination. Indeed, other researchers have discussed the idea that different groups (and cultures) construct shared meanings of time (Ancona, Okhuysen & Perlow, 2001). Boundary-spanning group members who share common, practiced approaches to time (i.e. meanings related to speed, punctuality, deadlines, industry rhythms, etc.)

should experience the same benefits mentioned above that are provided by other mental models. These benefits include the ability to draw inferences, to make predictions of future events (but not to regard those predictions as certainties), and to recognize relationships found in the environment. All these benefits directly relate to a group's shared understanding of its own and other timescapes, all of which denote a broader, richer temporal imagination being employed by the boundary spanning group. Likewise, a shared temporal mental model of practiced approaches to time should aid boundary spanning groups with heterogeneous compositions (informational or temporal) in tasks such as integrating information from varied perspectives, yet still allow a creative flow of different ideas and opinions during the integrative process.

Finally, we suggest that a shared temporal mental model within a group could act as an integrating mechanism, which we recommended earlier. [Ancona et al. \(2001\)](#) have described the need for such a mechanism (called "temporal coordination mechanisms") to control for differences in temporal perspectives and offset potential temporal conflicts between groups and outside, environmental entities. We suggest that a shared temporal mental model within a group could act as such a mechanism and could also make other such mechanisms easier to create and implement.

Group Longevity

Organizational groups vary in the length of time they exist; some groups come together for short, project-based tasks, then separate, while other groups are created for more strategic, long-term purposes. How does this difference affect the temporal characteristics of these groups, and is one form preferable over the other when boundary spanning is the group's purpose? We suggest that both short-term *and* long-term groups are appropriate, depending upon the reason for boundary spanning, and in fact, a balance of both types of groups might be preferable within organizations. In general, a combination of long-term and short-term boundary spanning groups helps create a balance of perspectives within organizations.

On the one hand, [Bluedorn and Ferris \(2000\)](#) found that the longer the organization has been in existence, the longer its temporal depth would be. Therefore, it seems reasonable to suggest that when organizations need a long-range perspective, stable, long-term boundary spanning groups should be emphasized. Such groups will gather and interpret information from the environment of interest for a longer period of time as a unified body, the result of which should be a clearer picture of how their timescape relates to the external one.

For groups that are interorganizational in nature, longer group longevity might be particularly important. Such groups are composed of members that represent

different (and often competitive) organizational parties. Longer tenure for these groups is necessary to ensure norms of reciprocity (Ziller, 1965), as well as feelings of trust and collaborative tendencies among members.

However, researchers also have shown the merits of short-term groups. One such virtue relates to the effect of familiarity. Researchers found that the positive effect of familiarity among group members is most significant early in the life of the group (Cannon-Bowers et al., 1995; Katz, 1982). Such familiarity allows groups to come up to speed quickly and coordinate rapidly. However, if the group remains intact over long periods of time, familiarity leads to a decrease in communication and in performance.³ One could speculate that familiarity might push the group toward groupthink in its deliberations and decision making too.

Given the apparent benefits of both long- and short-term groups, what is best in terms of temporal issues and boundary spanning groups? We suggest that groups of varying tenure be employed to engage in boundary spanning activities. In terms of the organization's comprehension and alignment of its timescape to that of its environment, deeper, richer understandings are created by taking advantage of the strengths of groups of varying tenure. Long-term groups provide long-term experience and perspective, while short-term groups provide flexibility and faster decision-making regarding domain-specific issues. If both short- and long-term groups are in place, the organization has a greater variety of perspectives and information it can draw upon, depending, of course, upon the reason and need for the boundary spanning activity. (We should note that references to either short- or long-term groups does not imply anything about how much time group members spend with each other in boundary spanning or other activities. As with many group activities in organizations, much group work will be carried out by people acting individually. Group activities conducted by the group acting as a face-to-face whole may only occur in meetings held a few times during the year.)

Degree of Autonomy

This final group attribute concerns the degree to which boundary spanning groups are allowed the freedom to direct their own activities. Groups with autonomy usually perform tasks that are highly interdependent within the group, and are usually given the responsibility and authority to control aspects of their work related to those tasks (Guzzo & Dickson, 1996). Given the nature of their goals, we feel boundary spanning groups benefit from a sense of autonomy, particularly from a temporal standpoint.

Such autonomy allows boundary spanning groups to create their own timescape, which may (or may not) mirror the timescapes of parent organizations; in fact,

the timescape of the boundary spanning group might actually be more similar to outside entities than to the organizations that created the groups. This can be a good thing. To achieve the goals of boundary spanning, these groups must be allowed to create a timescape that allows them to serve their parent organizations best, and such timescapes should reflect and integrate aspects of the larger social system with which these groups need to interact, and then interpret for their parent organizations. This is the very nature of boundary spanning. Furthermore, when the timescapes of these boundary spanning groups successfully reflect and integrate these elements, the boundary spanning group becomes a sort of lubricant between the organizations,⁴ allowing an easier flow of communication and information than might otherwise occur without an ongoing, connecting entity. Without the freedom and authority to break from temporal conformity with their organizations, boundary spanning groups will be restricted in their ability to accurately scan, assess, and represent the environment to the parent organizations, and likewise, the organizations back to the external environment.

CONCLUSION

In this chapter, we have proposed the concept of the temporal imagination, not as a precisely defined theoretical construct that can be instantly operationalized to begin a program of empirical research, but like its sociological progenitor, as a sensitizing concept (Blumer, 1954). A sensitizing concept is a general idea that can point to phenomena that are potentially important (Blumer, 1954, p. 7). In this case, it points to them for time and group scholars to consider, perhaps to investigate.

The results of such consideration may reveal significant competitive advantages to any individual, group, or organization that cultivates the temporal imagination's development and integrates it into strategic deliberations. Competitive advantages accrue not just because additional data are obtained and interpreted, as an information-processing perspective would suggest (e.g. Galbraith, 1977), but because a perspicacious temporal imagination that serves individuals, groups, and organizations well is likely to result from a confluence of idiosyncratic factors such as the histories of both the imaginers (i.e. individuals, groups, and organizations) and the timescapes of which they are a part. Such idiosyncrasies will make each temporal imagination unique; indeed, each application of each temporal imagination will be unique. Thus specific manifestations of temporal imaginations, being unique, will also be inimitable, and according to the resource-based theory of the firm, inimitability is a key attribute of any organizational resource that gives the firm a competitive advantage (Barney, 1986, 1991). And the more inimitable

the resource, the longer-lived the competitive advantage will be – something that is likely to be true for individuals and groups, just as it is for organizations.

Regardless of whether the temporal imagination invests its practitioners with significant competitive advantages vis-à-vis others, it promises to provide its practitioners with a better understanding of their worlds. This may be the most important advantage of all.

NOTES

1. [Bluedorn \(2002, pp. 278–279\)](#) has demonstrated that polychronicity was an attribute of the organizations he studied by illustrating the point that polychronicity can be a characteristic of both individuals and collectivities, the former having been demonstrated in several studies (see [Bluedorn, 2002, pp. 48–82](#)). In a similar vein, individuals can obviously develop and use temporal imaginations, and when acting collectively as in a boundary spanning group, such a group can *potentially* develop practices for understanding timescapes – the group's own as well as others. This could happen if the synergies of within-group interaction produce perceptions and understandings beyond those of any individual group member. In this sense a group can be described as developing and using a temporal imagination.

2. While the focus of this discussion is the effect of informational diversity upon the group's temporal development, we feel the positive effect of informational diversity (as well as the effect of some forms of temporal diversity, which we discuss below) upon the group's timescape and temporal imagination will ultimately have a positive effect upon the group's effectiveness for the organization as a boundary-spanning entity as well. In other words, we feel that the relationship between informational (and temporal) diversity and group effectiveness is an indirect one, mediated by the richness and complexity of the group's timescape and temporal imagination.

[Williams and O'Reilly \(1998\)](#) maintain that the impact of diversity in groups goes beyond simple main effects. Likewise, the assertion that the relationship between group diversity and performance is an indirect one has been studied by group researchers, with mixed results. For instance, previous researchers have found a direct link between diversity and effectiveness ([Bantel & Jackson, 1989](#); [Magjuka & Baldwin, 1991](#)); however, recent studies have found no such direct correlation ([Jehn et al., 1999](#)), stressing instead the importance of mediating constructs such as task conflict. Other researchers have suggested such mediating constructs as task complexity and the interdependence of group members. We feel that because we know the purpose of our group (boundary spanning), we can make some assumptions regarding the nature and context of these other mediating variables. Likely boundary-spanning tasks include gathering and assessing environmental information for the purposes of organizational decision making and strategizing. Such tasks are typically complex, ambiguous, and require a fair amount of interdependence among the group members. Therefore, our discussions of information and temporal diversity and their impact upon groups presumes such a task environment.

3. It is worth considering that the decrement in communication and performance in the above studies might also come from other intervening constructs – see our previous note about the impact of mediating variables upon group outcomes.

4. We would like to thank editor Margaret Neale for suggesting to us the concept of the boundary spanning group as lubricant.

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REFERENCES

- Adam, B. (1998). *Timescapes of modernity: The environment and invisible hazards*. London: Routledge.
- Ancona, D., & Chong, C. L. (1996). Entrainment: Pace, cycle, and rhythm in organizational behavior. *Research in Organizational Behavior*, 18, 251–284.
- Ancona, D. G., & Caldwell, D. F. (1992). Demography and design: Predictors of new product team performance. *Organization Science*, 3, 321–341.
- Ancona, D. G., Okhuysen, G. A., & Perlow, L. A. (2001). Taking time to integrate temporal research. *Academy of Management Review*, 26, 512–529.
- Bantel, K. A., & Jackson, S. E. (1989). Top management and innovations in banking: Does composition of the top teams make a difference? *Strategic Management Journal*, 10(Special Issue, Summer), 107–124.
- Barney, J. (1991). Firm resources and sustained competitive advantage. *Journal of Management*, 17, 99–120.
- Barney, J. B. (1986). Organizational culture: Can it be a source of sustained competitive advantage? *Academy of Management Review*, 11, 656–665.
- Barth, J. R. (1991). The temporal imagination in Wordsworth's *Prelude*: Time and the timeless. *Thought*, 66(261), 139–150.
- Benford, G. (1992). *Timescape*. New York: Bantam Books. (Original work published in 1980.)
- Blackburn, J. D. (Ed.) (1991). *Time-based competition: The next battleground in American manufacturing*. Homewood, IL: Business One Irwin.
- Bluedorn, A. C. (1997). Primary rhythms, information processing, and planning: Toward a strategic temporal technology. *Technology Studies*, 4, 1–36.
- Bluedorn, A. C. (1998). An interview with anthropologist Edward T. Hall. *Journal of Management Inquiry*, 7, 109–115.
- Bluedorn, A. C. (2000). Time and organizational culture. In: N. M. Ashkanasy, C. P. M. Wilderom & M. F. Peterson (Eds), *Handbook of Organizational Culture and Climate* (pp. 117–128). Thousand Oaks, CA: Sage.
- Bluedorn, A. C. (2002). *The human organization of time: Temporal realities and experience*. Stanford, CA: Stanford University Press.
- Bluedorn, A. C., & Ferris, S. P. (2000, October 13). Temporal depth, age, and organizational performance. Paper presented in the INSEAD Strategic Management Seminar Series, Fontainebleau, France.

- Bluedorn, A. C., Kaufman, C. F., & Lane, P. M. (1992). How many things do you like to do at once? An introduction to monochronic and polychronic time. *Academy of Management Executive*, 6(4), 17–26.
- Bluedorn, A. C., & Lundgren, E. F. (1993). A culture-match perspective for strategic change. *Research in Organizational Change and Development*, 7, 137–179.
- Blumer, H. (1954). What is wrong with social theory. *American Sociological Review*, 19, 3–10.
- Cannon-Bowers, J. A., Salas, E., & Converse, S. (1993). Shared mental models in expert team decision making. In: N. J. Castellan, Jr. (Ed.), *Individual and Group Decision Making* (pp. 221–246). Hillsdale, NJ: Lawrence Erlbaum.
- Cannon-Bowers, J. A., Tannenbaum, S. I., Salas, E., & Volpe, C. E. (1995). Defining competencies and establishing team training requirements. In: R. A. Guzzo & E. Salas (Eds), *Team Effectiveness and Decision Making in Organizations* (pp. 333–380). San Francisco, CA: Jossey-Bass.
- Chowers, E. (1998). Time in Zionism: The life and afterlife of a temporal revolution. *Political Theory*, 26, 652–685.
- De Dreu, C. K. W., & Weingart, L. R. (2002, August 9–14). Task versus relationship conflict: A meta-analysis. Paper presented at the Annual Meeting of the Academy of Management, Denver, CO.
- DiMaggio, P. J., & Powell, W. W. (1983). The iron cage revisited: Institutional isomorphism and collective rationality in organizational fields. *American Sociological Review*, 48, 147–160.
- Eisenhardt, K. M. (1989). Making fast strategic decisions in high-velocity environments. *Academy of Management Journal*, 32, 543–576.
- Galbraith, J. R. (1977). *Organization design*. Reading, MA: Addison-Wesley.
- Greening, D. W., & Johnson, R. A. (1997). Managing industrial and environmental crises. *Business and Society*, 36, 334–361.
- Guzzo, R. A., & Dickson, M. W. (1996). Teams in organizations: Recent research on performance and effectiveness. *Annual Review of Psychology*, 47, 307–338.
- Hall, E. T. (1981). *The silent language*. New York: Anchor Books. (Original work published in 1959.)
- Hall, E. T. (1983). *The dance of life: The other dimension of time*. Garden City, NY: Anchor Press.
- Hambrick, D. C., Cho, T. S., & Chen, M.-J. (1996). The influence of top management team heterogeneity on firm's competitive moves. *Administrative Science Quarterly*, 41, 659–684.
- Jackson, S. (1992). Team composition in organizational settings: Issues in managing an increasingly diverse work force. In: S. Worchel, W. Wood & J. A. Simpson (Eds), *Group Process and Productivity* (pp. 138–173). London: Sage.
- Jackson, S. E., May, K. E., & Whitney, K. (1995). Understanding the dynamics of diversity in decision-making teams. In: R. A. Guzzo & E. Salas (Eds), *Team Effectiveness and Decision Making in Organizations* (pp. 204–261). San Francisco, CA: Jossey-Bass.
- Jehn, K. A., Northcraft, G. B., & Neale, M. A. (1999). Why differences make a difference: A field study of diversity, conflict, and performance in workgroups. *Administrative Science Quarterly*, 44, 741–763.
- Kahn, R. L., Wolfe, D. M., Quinn, R. P., Snoek, J. D., & Rosenthal, R. A. (1964). *Organizational stress: Studies in role conflict and ambiguity*. New York: Wiley.
- Katz, R. (1982). The effects of group longevity on project communication and performance. *Administrative Science Quarterly*, 27, 81–104.
- Klimoski, R., & Mohammed, S. (1994). Team mental model: Construct or metaphor? *Journal of Management*, 20, 403–437.
- Kluckhohn, F. R., & Strodtbeck, F. L. (1961). *Variations in value orientations*. Evanston, IL: Row, Peterson.

- Lau, D. C., & Murnighan, J. K. (1998). Demographic diversity and faultlines: The compositional dynamics of organizational groups. *Academy of Management Review*, 23, 325–340.
- Lawrence, P. R., & Lorsch, J. W. (1967). *Organization and environment: Managing differentiation and integration*. Boston, MA: Division of Research, Graduate School of Business Administration, Harvard University.
- Levine, R. (1997). *The geography of time: The temporal misadventures of a social psychologist, or how every culture keeps time just a little bit differently*. New York: Basic Books.
- Levine, R. V., West, L. J., & Reis, H. T. (1980). Perceptions of time and punctuality in the United States and Brazil. *Journal of Personality and Social Psychology*, 38, 541–550.
- Levine, R., & Wolff, E. (1985). Social time: The heartbeat of culture. *Psychology Today*, 19(3), 28–35.
- Likert, R. (1961). *New patterns of management*. New York: McGraw-Hill.
- Magjuka, R. J., & Baldwin, T. T. (1991). Team-based employee involvement programs: Effects of design and administration. *Personnel Psychology*, 44, 793–812.
- March, J. G., & Simon, H. A. (1958). *Organizations*. New York: Wiley.
- Mathieu, J. E., Heffner, T. S., Goodwin, G. F., Salas, E., & Cannon-Bowers, J. A. (2000). The influence of shared mental models on team process and performance. *Journal of Applied Psychology*, 85, 273–283.
- Mead, G. H. (1974). *Mind, self, and society: From the standpoint of a social behaviorist*. Chicago: University of Chicago Press. (Original work published 1934.)
- Mills, C. W. (1959). *The sociological imagination*. London: Oxford University Press.
- Morgan, G. (1997). *Images of organization* (2nd ed.). Thousand Oaks, CA: Sage.
- Nock, S. L. (1998). Turn-taking as rational behavior. *Social Science Research*, 27, 235–244.
- Ofori-Dankwa, J., & Julian, S. D. (2001). Complexifying organizational theory: Illustrations using time research. *Academy of Management Review*, 26, 415–430.
- Pennings, J. M. (1998). Structural contingency theory. In: P. J. D. Drenth & C. J. de Wolff (Eds), *Handbook of Work and Organizational Psychology* (pp. 39–60). East Sussex, UK: Psychology Press Ltd.
- Richards, E. G. (1998). *Mapping time: The calendar and its history*. Oxford: Oxford University Press.
- Saxe, J. G. (1892). *The poetical works of John Godfrey Saxe*. Boston: Houghton, Mifflin and Company.
- Schein, E. H. (1992). *Organizational culture and leadership* (2nd ed.). San Francisco: Jossey-Bass.
- Stalk, G., Jr., & Hout, T. M. (1990). *Competing against time: How time-based competition is reshaping global markets*. New York: Free Press.
- Stout, R. J., Cannon-Bowers, J. A., & Salas, E. (1996). The role of shared mental models in developing team situational awareness: Implications for training. *Training Research Journal*, 2, 85–116.
- Thompson, J. D. (1967). *Organizations in action*. New York: McGraw-Hill.
- Weitzel, W. C., III (1999). *The space of memory: Romanticism, modernity, and the temporal imagination*. Unpublished doctoral dissertation, Harvard University, Cambridge, MA.
- Whitrow, G. J. (1988). *Time in history: The evolution of our general awareness of time and temporal perspective*. Oxford: Oxford University Press.
- Williams, K. Y., & O'Reilly, C. A. (1998). Demography and diversity in organizations: A review of forty years of research. *Research in Organizational Behavior*, 20, 77–140.
- Ziller, R. C. (1965). Toward a theory of open and closed groups. *Psychological Bulletin*, 64, 164–182.